

4. Project Planning Phase

ComicCraft – AI Comic Story Creator Using Gemini Models

Organizing milestones, activities, dependencies, and deliverables

1. Planning Objective

The project guide organizes implementation into milestones. This phase presents those milestones as a practical sequence for completing and verifying ComicCraft.

2. Milestone 1 – Model Selection and Architecture

Activities: research and select generative AI models; define the application architecture; set up the development environment. Deliverables: selected model roles, architecture outline, and a prepared project environment.

3. Milestone 2 – Core Functionality Development

Activities: implement outline generation, story generation, image generation, layout building, and PDF export; implement the FastAPI backend for routing and user-input processing. Deliverables: core generation modules and backend integration.

4. Milestone 3 – routes.py Development

Activity: implement the main application logic and route handlers in routes.py. Deliverable: routes that connect user requests to the comic-generation workflow and handle failures.

5. Milestone 4 – Frontend Development

Activities: design and develop the user interface; create dynamic templates with FastAPI's Jinja2 integration. Deliverables: homepage, comic preview, and export-success pages.

6. Milestone 5 – Deployment and Local Verification

Activities: prepare the application for local deployment; run and verify it locally. Deliverables: configured environment, running local server, and verified browser/API access.

7. Dependencies and Sequence

Model and architecture decisions inform module implementation. Core functions are then connected through routes. The frontend templates display the backend results. Local deployment and verification follow integration of the application components.

8. Planning Considerations

Prepare the Python environment and dependencies; configure Gemini and Hugging Face credentials through environment variables; ensure static and template directories are available; and reserve time to test both browser and API workflows.

9. Planning Outcome

The project plan provides a milestone-based implementation path from initial setup through core functionality, route integration, frontend development, and local verification.